

Intensity Transformations and Geometric Transformations

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Image Processing

- Interpolation
- Intensity Transformations
- Geometric Transformations



INTERPOLATION

Interpolation

Interpolation estimates unknown values from known data points, crucial for image resizing and geometric transformations. For example, to enlarge a 500x500 image to 750x750:

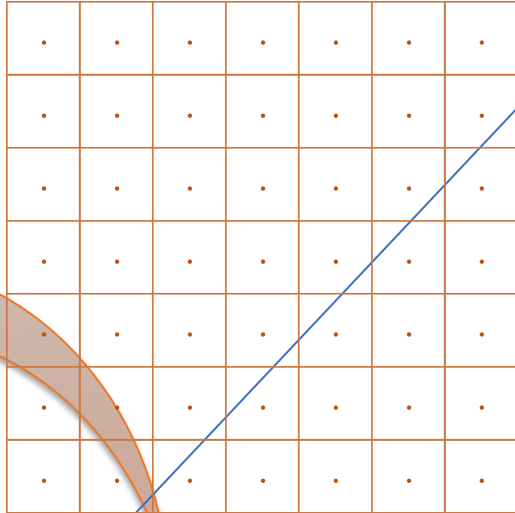
1. Create a 750x750 grid
2. Shrink this grid to match original image size
3. Assign intensity values by matching each new grid point to nearest original pixel (nearest neighbor interpolation)

Interpolation ใช้ในการประมาณค่าที่ไม่รู้จากข้อมูลที่อยู่ จำเป็นสำหรับการปรับขนาดและแปลงทางเรขาคณิต ตัวอย่างเช่น การขยายภาพจาก 500 x 500 เป็น 750 x 750 :

1. สร้างตารางขนาด 750 x 750
2. หดตารางนี้ให้ตรงกับขนาดภาพต้นฉบับ
3. กำหนดค่าความเข้มโดยจับคู่จุดตารางใหม่กับพิกเซลต้นฉบับที่ใกล้ที่สุด

Nearest Neighbor Interpolation

0	0	0	0	0
0	0.7	0.8	0.3	0
0	0.8	0	0.9	0
0	0.2	0	1	0
0	1	0	0.8	0



0	0	0	0	0	0	0
0	0.7	0.7	0.8	0.3	0.3	0
0	0.7	0.7	0.8	0.3	0.3	0
0	0.8	0.8	0	0.9	0.9	0
0	0.2	0.2	0	1	1	0
0	0.2	0.2	0	1	1	0
0	1	1	0	0.8	0.8	0

0	0	0	0	0
0	0.7	0.8	0.3	0
0	0.8	0	0.9	0
0	0.2	0	1	0
0	1	0	0.8	0

0	0	0	0	0
0	0.7	0.8	0.3	0
0	0.8	0	0.9	0
0	0.2	0	1	0
0	1	0	0.8	0

Interpolation

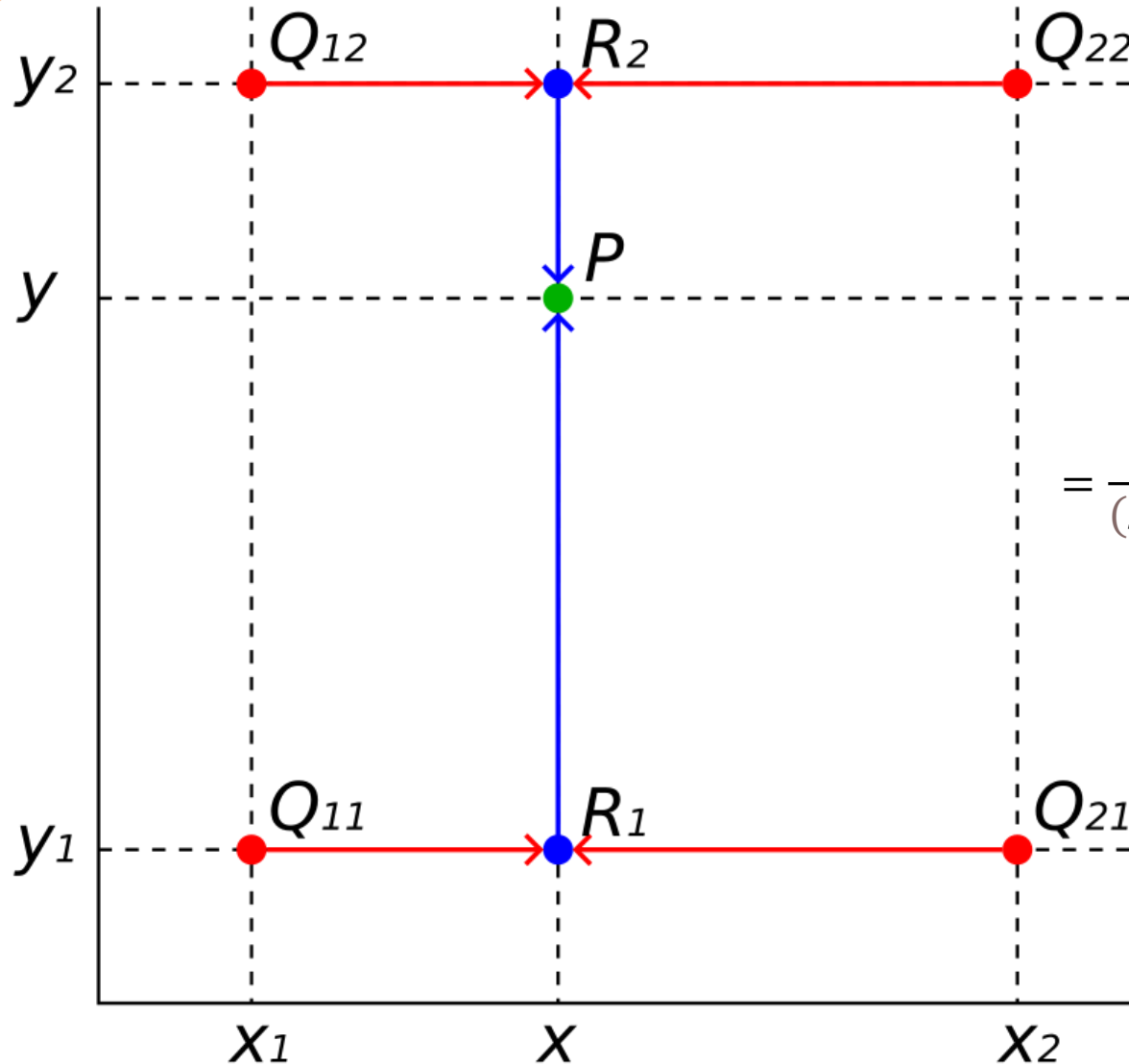
Nearest neighbor interpolation assigns new pixels the value of their closest original pixel. While simple, this can distort straight edges. Bilinear interpolation offers better results by:

1. Using four nearest neighbors instead of one
2. Calculating weighted average based on distance

การประมาณค่าแบบเพื่อนบ้านที่ใกล้ที่สุดกำหนดค่าพิกเซลใหม่จากพิกเซลต้นฉบับที่ใกล้ที่สุด แม้จะง่าย แต่อาจทำให้ขอบที่เคยตรงผิดเพี้ยนได้ การประมาณค่าเชิงเส้นแบบคู่ (Bilinear interpolation) ให้ผลลัพธ์ที่ดีกว่าโดย:

1. ใช้เพื่อนบ้านที่ใกล้ที่สุดสี่จุด
2. คำนวณค่าเฉลี่ยถ่วงน้ำหนักตามระยะทาง

Bilinear interpolation



$$f(R_2) = f(x, y_2) = \frac{x_2 - x}{x_2 - x_1} f(Q_{12}) + \frac{x - x_1}{x_2 - x_1} f(Q_{22})$$

$$f(P) = f(x, y) = \frac{y_2 - y}{y_2 - y_1} f(R_2) + \frac{y - y_1}{y_2 - y_1} f(R_1)$$

$$= \frac{1}{(x_2 - x_1)(y_2 - y_1)} \begin{bmatrix} x_2 - x & x - x_1 \end{bmatrix} \begin{bmatrix} f(Q_{11}) & f(Q_{12}) \\ f(Q_{21}) & f(Q_{22}) \end{bmatrix} \begin{bmatrix} y_2 - y \\ y - y_1 \end{bmatrix}$$

$$f(R_1) = f(x, y_1) = \frac{x_2 - x}{x_2 - x_1} f(Q_{11}) + \frac{x - x_1}{x_2 - x_1} f(Q_{21})$$

Bilinear interpolation

0	0	0	0	0
0	0.7	0.8	0.3	0
0	0.8	0	0.9	0
0	0.2	0	1	0
0	1	0	0.8	0

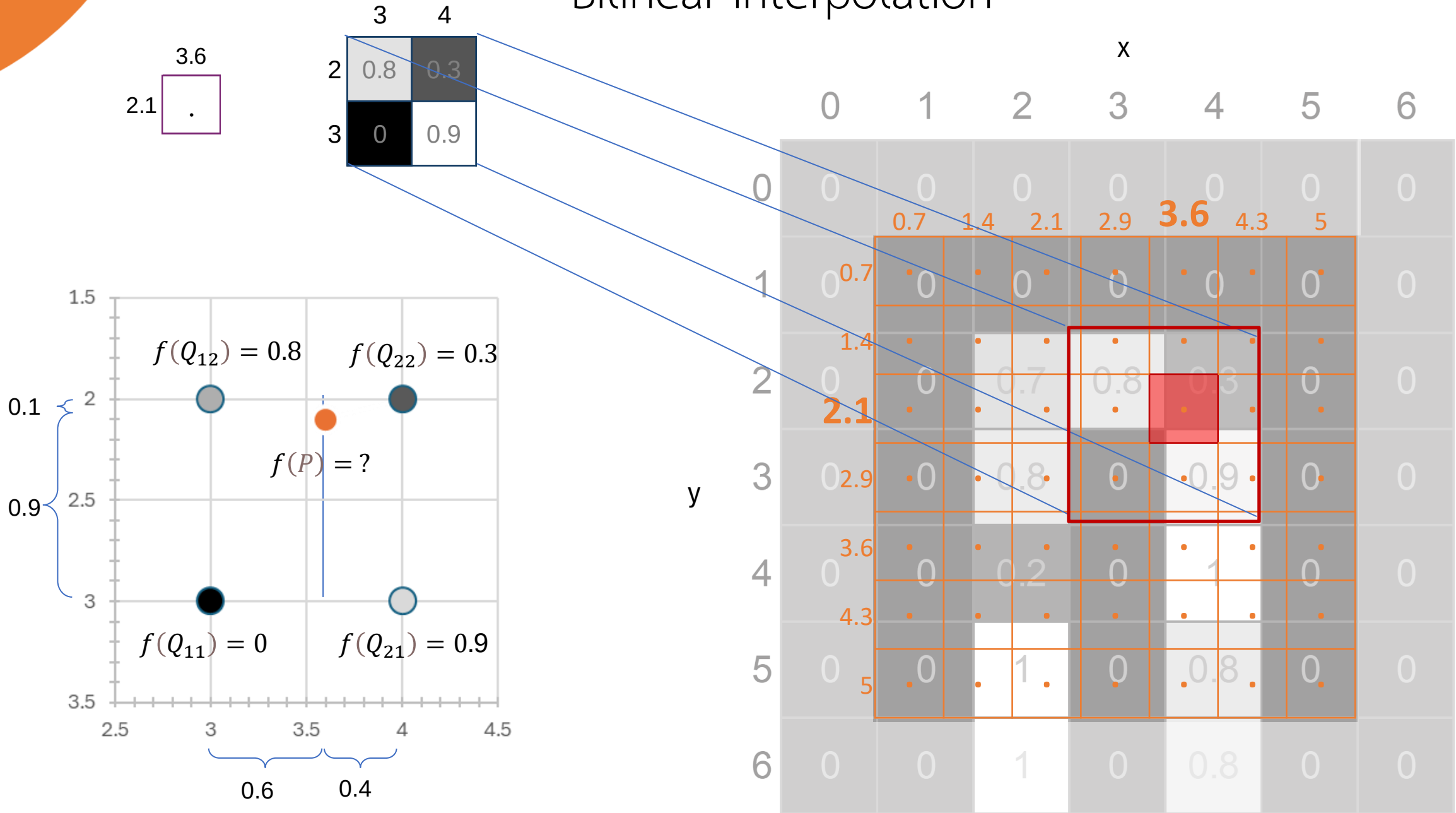
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.

0	0	0	0	0	0	0
0	0.1	0.3	0.3	0.2	0.1	0
0	0.3	0.7	0.7	0.5	0.3	0
0	0.1	0.2	0	0.6	0.7	0
0	0.3	0.6	0.1	0.5	0.6	0
0	0.4	0.9	0.1	0.5	0.6	0
0	0.4	0.9	0.1	0.5	0.6	0

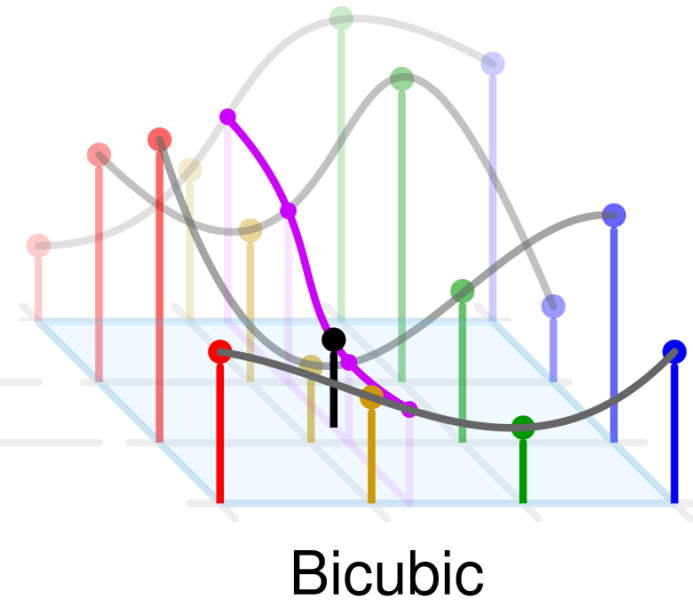
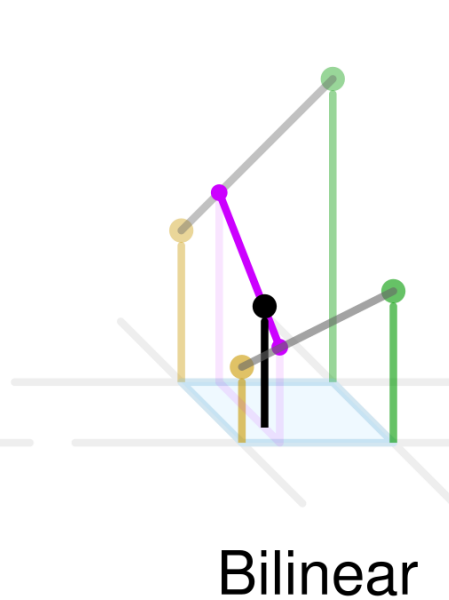
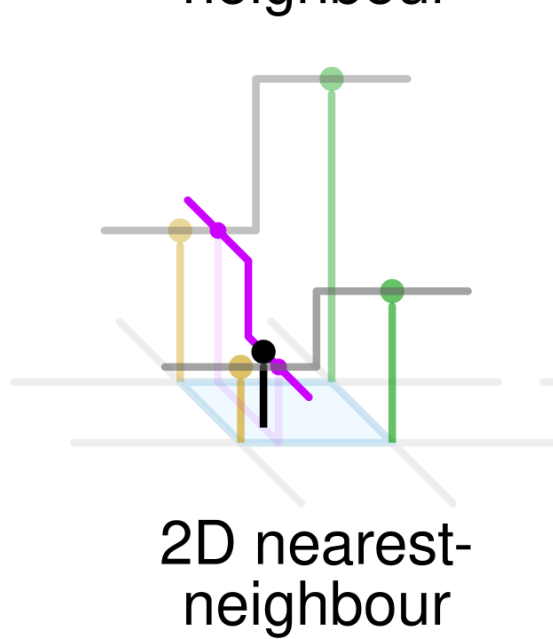
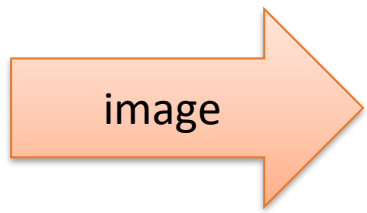
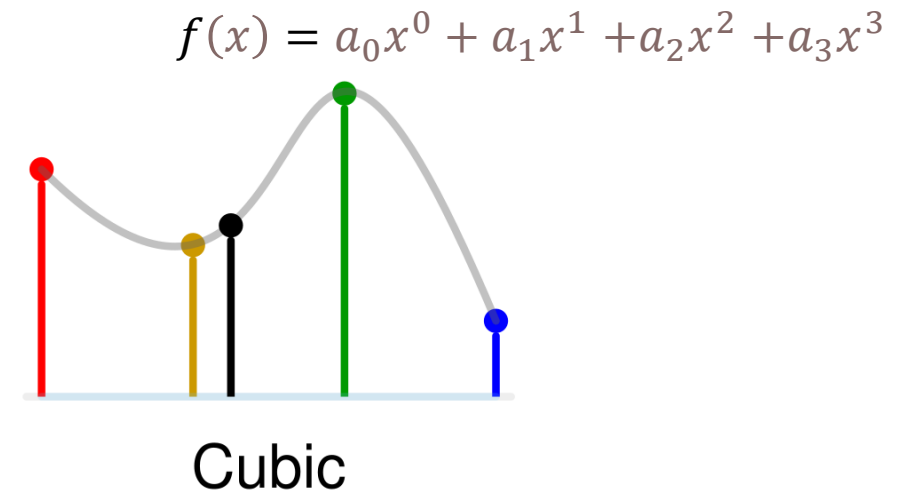
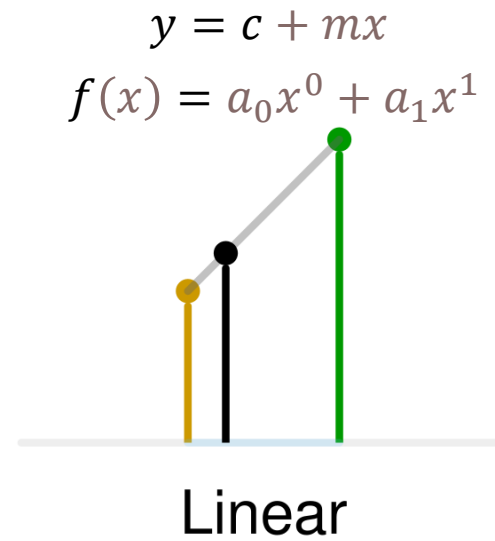
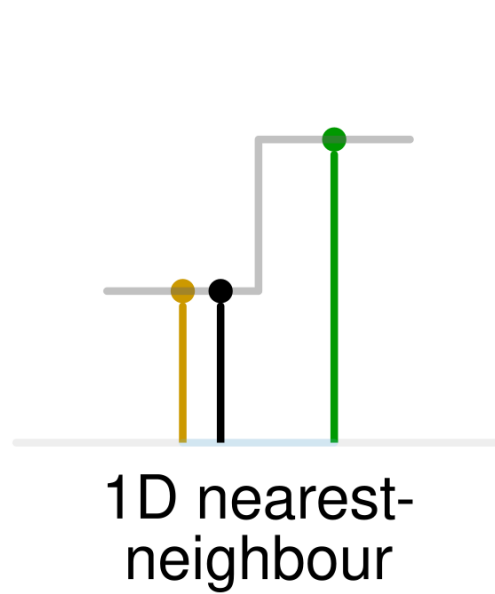
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0.7	0.8	0.3	0	0
0	0	0.8	0	0.9	0	0
0	0	0.2	0	1	0	0
0	0	1	0	0.8	0	0
0	0	1	0	0.8	0	0

	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0 ^{0.7}	0	0	0	0	0	0
2	0 ^{1.4}	0	0.7	0.8	0.3	0	0
3	0 ^{2.1}	0	0.8	0	0.9	0	0
4	0 ^{2.9}	0	0.2	0	1	0	0
5	0 ^{3.6}	0	1	0	0.8	0	0
6	0 ^{4.3}	0	1	0	0.8	0	0

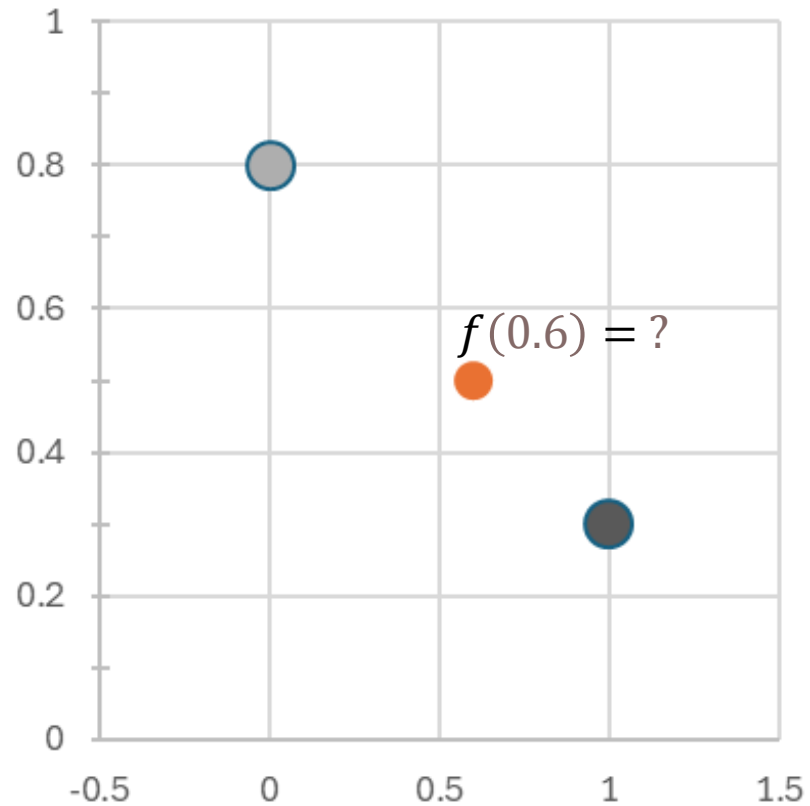
Bilinear interpolation



Interpolation in general form



linear interpolation



$$f(x) = a_0x^0 + a_1x^1$$

$$f(0) = 0.8 = a_0 * 1 + a_1 * 0$$

$$f(1) = 0.3 = a_0 * 1 + a_1 * 1$$

$$\mathbf{f} = \mathbf{X}\mathbf{a}$$

$$\begin{bmatrix} f(0) \\ f(1) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix}$$

$$\mathbf{a} = \mathbf{X}^{-1}\mathbf{f}$$

$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} f(0) \\ f(1) \end{bmatrix}$$

Recycle matrix

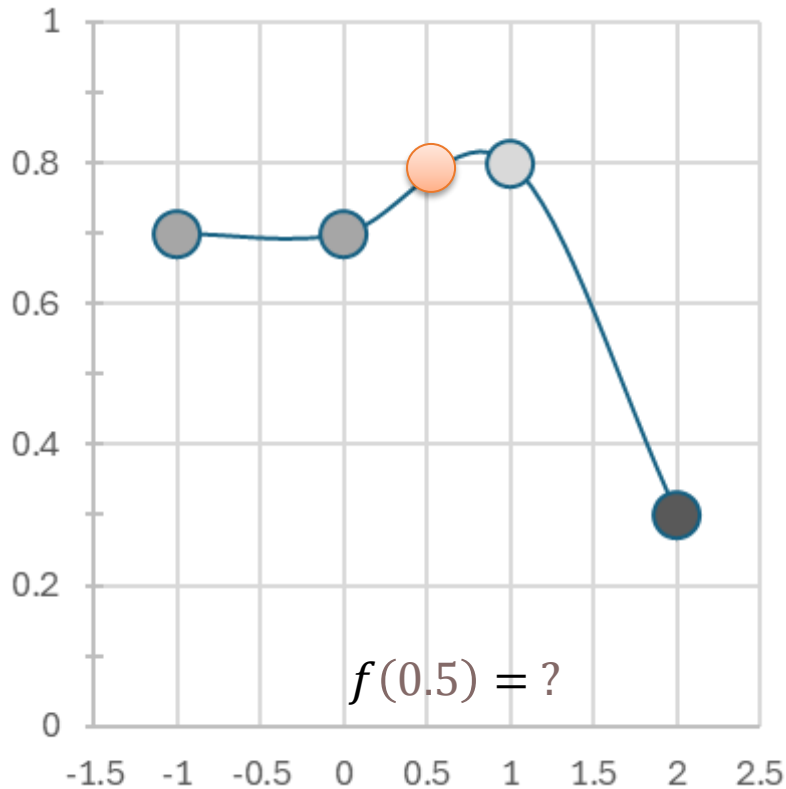
$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 0.8 \\ -0.5 \end{bmatrix}$$

$$f(x) = 0.8 - 0.5x$$

$$f(0.6) = 0.5$$

Cubic interpolation

$$f(x) = a_0x^0 + a_1x^1 + a_2x^2 + a_3x^3$$



$$f(-1) = 0.7 = a_0 - a_1 + a_2 - a_3$$

$$f(0) = 0.7 = a_0 + a_2$$

$$f(1) = 0.8 = a_0 + a_1 + a_2 + a_3$$

$$f(2) = 0.3 = a_0 + 2a_1 + 4a_2 + 8a_3$$

$$\mathbf{f} = \mathbf{X}\mathbf{a}$$

$$\begin{bmatrix} f(-1) \\ f(0) \\ f(1) \\ f(2) \end{bmatrix} = \begin{bmatrix} 1 & -1 & 1 & -1 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

Recycle matrix

$$\mathbf{a} = \mathbf{X}^{-1}\mathbf{f}$$

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 0 & 6 & 0 & 0 \\ -2 & -3 & 6 & -1 \\ 3 & -6 & 3 & 0 \\ -1 & 3 & -3 & 1 \end{bmatrix} \begin{bmatrix} f(-1) \\ f(0) \\ f(1) \\ f(2) \end{bmatrix}$$

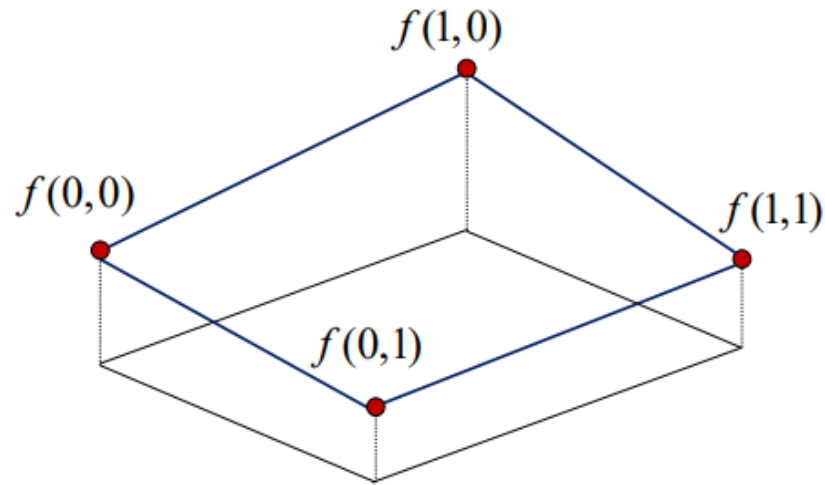
$$f(x) = 0.7 + 0.167x^1 + 0.05x^2 - 0.117x^3$$

$$f(0.5) = 0.78$$

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 0.7 \\ 0.167 \\ 0.05 \\ -0.117 \end{bmatrix}$$

Bilinear interpolation

Find the equation of the plane between points.



$$f(x, y) = \sum_{j=0}^1 \sum_{i=0}^1 a_{ij} x^i y^j$$

$$= a_{00} + a_{10}x + a_{01}y + a_{11}xy$$

$$x, y \in \{-1, 0, 1, 2\}$$

System of Equations $y = Xa$

$$\begin{bmatrix} f(0,0) \\ f(1,0) \\ f(0,1) \\ f(1,1) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} a_{00} \\ a_{10} \\ a_{01} \\ a_{11} \end{bmatrix}$$

Solution $a = X^{-1}y$

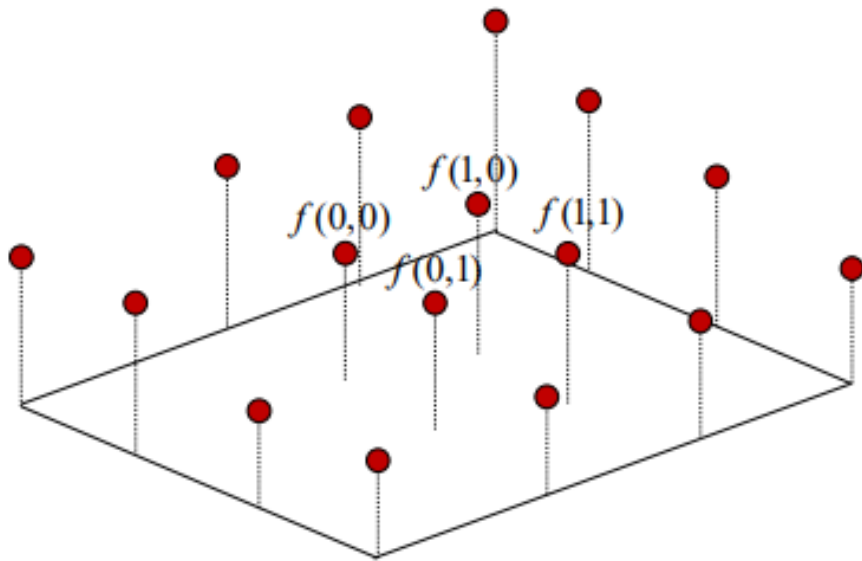
$$\begin{bmatrix} a_{00} \\ a_{10} \\ a_{01} \\ a_{11} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 1 & -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} f(0,0) \\ f(1,0) \\ f(0,1) \\ f(1,1) \end{bmatrix}$$



recycle matrix

BiCubic Interpolation

System of Equations: 16 equations, 16 unknowns



$$f(x, y) = \sum_{j=0}^3 \sum_{i=0}^3 a_{ij} x^i y^j$$

$$= a_{00} + a_{10}x + a_{01}y + a_{11}xy$$

$$x, y \in \{-1, 0, 1, 2\}$$

BiCubic Interpolation

System of Equations: 16 equations, 16 unknowns

$$y = Xa$$

$$\begin{bmatrix} f(-1,-1) \\ f(0,-1) \\ f(1,-1) \\ f(2,-1) \\ f(-1,0) \\ f(0,0) \\ f(1,0) \\ f(2,0) \\ f(-1,1) \\ f(0,1) \\ f(1,1) \\ f(2,1) \\ f(-1,2) \\ f(0,2) \\ f(1,2) \\ f(2,2) \end{bmatrix} = \begin{bmatrix} 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 & 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \\ 1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & 2 & 4 & 8 & -1 & -2 & -4 & -8 & 1 & 2 & 4 & 8 & -1 & -2 & -4 & -8 \\ 1 & -1 & 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 2 & 4 & 8 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 & 1 & 2 & 4 & 8 & 1 & 2 & 4 & 8 & 1 & 2 & 4 & 8 \\ 1 & -1 & 1 & -1 & 2 & -2 & 2 & -2 & 4 & -4 & 4 & -4 & 8 & -8 & 8 & -8 \\ 1 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 4 & 0 & 0 & 0 & 8 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 & 4 & 4 & 4 & 4 & 8 & 8 & 8 & 8 \\ 1 & 2 & 4 & 8 & 2 & 4 & 8 & 16 & 4 & 8 & 16 & 32 & 8 & 16 & 32 & 64 \end{bmatrix} \begin{bmatrix} a_{00} \\ a_{10} \\ a_{20} \\ a_{30} \\ a_{01} \\ a_{11} \\ a_{21} \\ a_{31} \\ a_{02} \\ a_{12} \\ a_{22} \\ a_{32} \\ a_{03} \\ a_{13} \\ a_{23} \\ a_{33} \end{bmatrix}$$

$$a = X^{-1}I$$

$$\begin{bmatrix} a_{00} \\ a_{10} \\ a_{20} \\ a_{30} \\ a_{01} \\ a_{11} \\ a_{21} \\ a_{31} \\ a_{02} \\ a_{12} \\ a_{22} \\ a_{32} \\ a_{03} \\ a_{13} \\ a_{23} \\ a_{33} \end{bmatrix} = \frac{1}{36} \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 36 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -12 & -18 & 36 & -6 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 18 & -36 & 18 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -6 & 18 & -18 & 6 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -12 & 0 & 0 & 0 & -18 & 0 & 0 & 0 & 36 & 0 & 0 & 0 & -6 & 0 & 0 \\ 4 & 6 & -12 & 2 & 6 & 9 & -18 & 3 & -12 & -18 & 36 & -6 & 2 & 3 & -6 & 1 \\ -6 & 12 & -6 & 0 & -9 & 18 & -9 & 0 & 18 & -36 & 18 & 0 & -3 & 6 & -3 & 0 \\ 2 & -6 & 6 & -2 & 3 & -9 & 9 & -3 & -6 & 18 & -18 & 6 & 1 & -3 & 3 & -1 \\ 0 & 18 & 0 & 0 & 0 & -36 & 0 & 0 & 0 & 18 & 0 & 0 & 0 & 0 & 0 & 0 \\ -6 & -9 & 18 & -3 & 12 & 18 & -36 & 6 & -6 & -9 & 18 & -3 & 0 & 0 & 0 & 0 \\ 9 & -18 & 9 & 0 & -18 & 36 & -18 & 0 & 9 & -18 & 9 & 0 & 0 & 0 & 0 & 0 \\ -3 & 9 & -9 & 3 & 6 & -18 & 18 & -6 & -3 & 9 & -9 & 3 & 0 & 0 & 0 & 0 \\ 0 & -6 & 0 & 0 & 0 & 18 & 0 & 0 & 0 & -18 & 0 & 0 & 0 & 6 & 0 & 0 \\ 2 & 3 & -6 & 1 & -6 & -9 & 18 & -3 & 6 & 9 & -18 & 3 & -2 & -3 & 6 & -1 \\ -3 & 6 & -3 & 0 & 9 & -18 & 9 & 9 & -9 & 18 & -9 & 0 & 3 & -6 & 3 & 0 \\ 1 & -3 & 3 & -1 & -3 & 9 & -9 & 3 & 3 & -9 & 9 & -3 & -1 & 3 & -3 & 1 \end{bmatrix} \begin{bmatrix} I(-1,-1) \\ I(0,-1) \\ I(1,-1) \\ I(2,-1) \\ I(-1,0) \\ I(0,0) \\ I(1,0) \\ I(2,0) \\ I(-1,1) \\ I(0,1) \\ I(1,1) \\ I(2,1) \\ I(-1,2) \\ I(0,2) \\ I(1,2) \\ I(2,2) \end{bmatrix}$$

↑
recycle matrix



INTENSITY TRANSFORMATION

Spatial Domain

The spatial domain refers to working directly with image pixels.

Spatial Operations work directly with pixels and come in three types:

1. Operations on individual pixels
2. Operations involving surrounding pixels
3. Geometric transformations of the image space

โดเมนเชิงพื้นที่หมายถึงการทำงานกับพิกเซลของภาพโดยตรง

การดำเนินการเชิงพื้นที่ที่ทำงานโดยตรงกับพิกเซลของภาพ แบ่งเป็นสามประเภท:

1. การดำเนินการกับพิกเซลแต่ละจุด
2. การดำเนินการกับพิกเซลรอบข้าง
3. การแปลงรูปทรงเรขาคณิตของพื้นที่ภาพ

Spatial Operations

- Single-Pixel Operations

$$g(x, y) = T[f(x, y)]$$

or $s = T(r)$

- Neighborhood Operations (average)

$$g(x, y) = \frac{1}{nm} \sum_{(r, c) \in S_{xy}} f(r, c)$$

Set of coordinates of pixels
in neighborhood region

- Geometric transformations of the image space

— consist of two basic operations:

- Spatial transformation of coordinates.
- Intensity interpolation that assigns intensity values to the spatially transformed pixels.

a b

FIGURE 3.2

Intensity transformation functions. (a) Contrast stretching function. (b) Thresholding function.

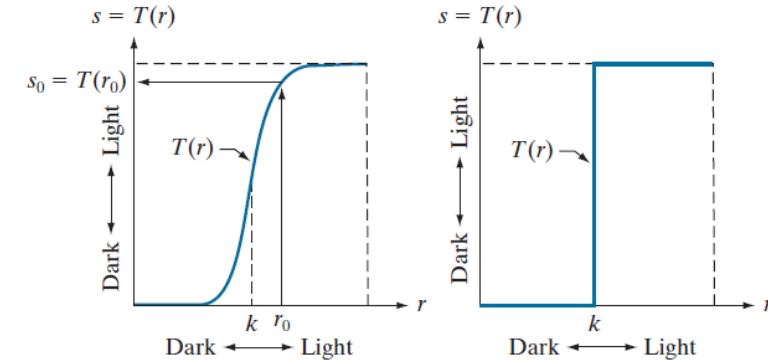
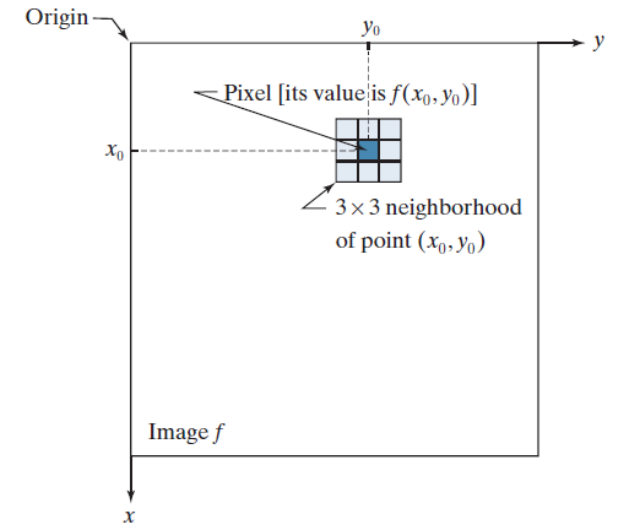


FIGURE 3.1

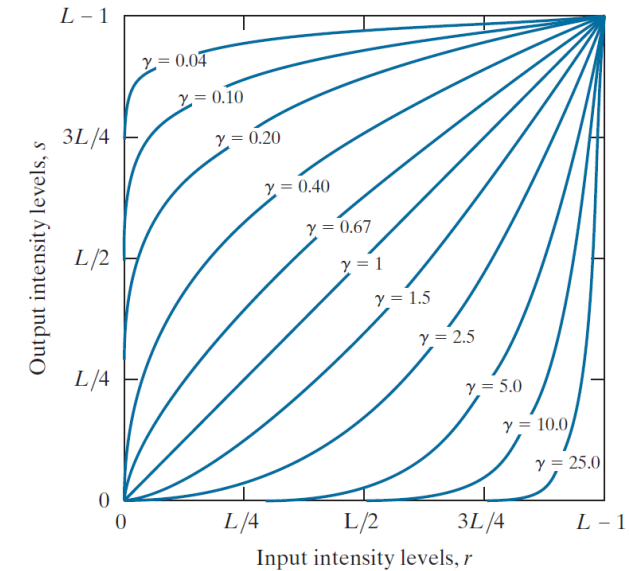
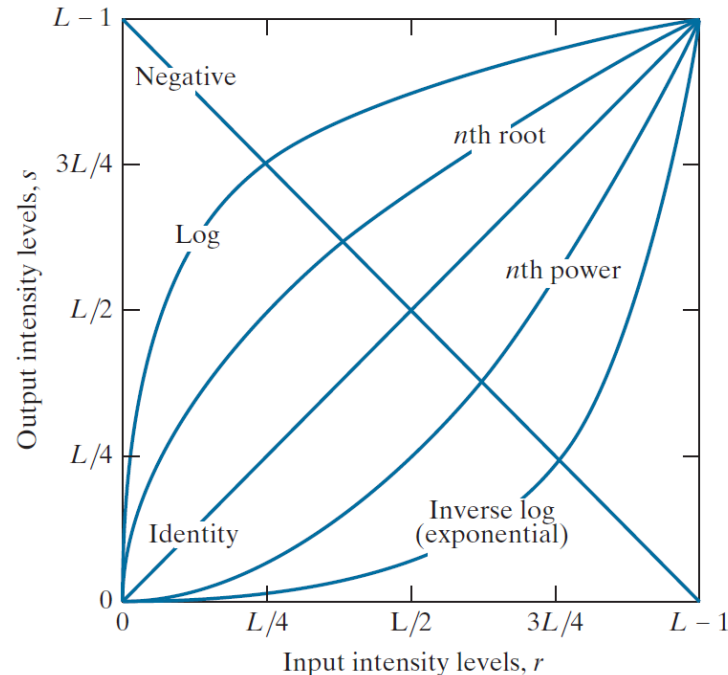
A 3×3 neighborhood about a point (x_0, y_0) in an image. The neighborhood is moved from pixel to pixel in the image to generate an output image. Recall from Chapter 2 that the value of a pixel at location (x_0, y_0) is $f(x_0, y_0)$, the value of the image at that location.



Some Basic Intensity Transformation Functions

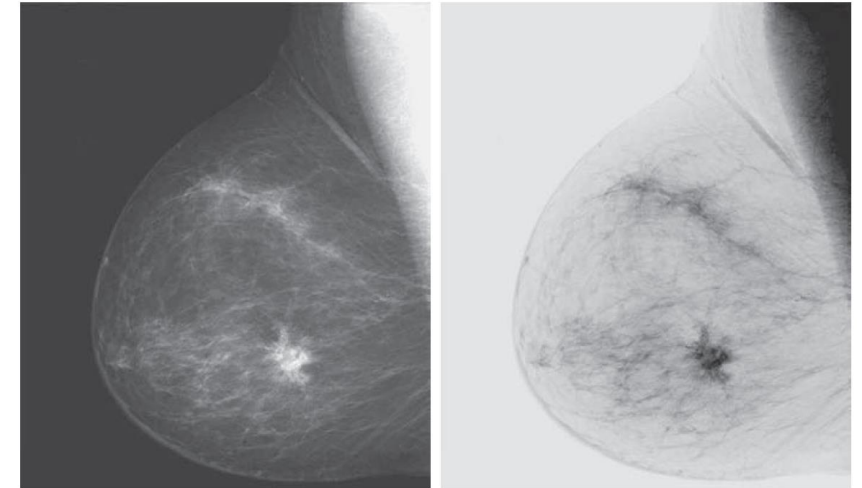
- Log Transformations $s = c \log(1 + r)$, $r \geq 0$.
- Power-law (Gamma) Transformations $s = cr^\gamma$

FIGURE 3.3
Some basic intensity transformation functions. Each curve was scaled *independently* so that all curves would fit in the same graph. Our interest here is on the *shapes* of the curves, not on their relative values.



a b

FIGURE 3.4
(a) A digital mammogram.
(b) Negative image obtained using Eq. (3-3).
(Image (a) Courtesy of General Electric Medical Systems.)



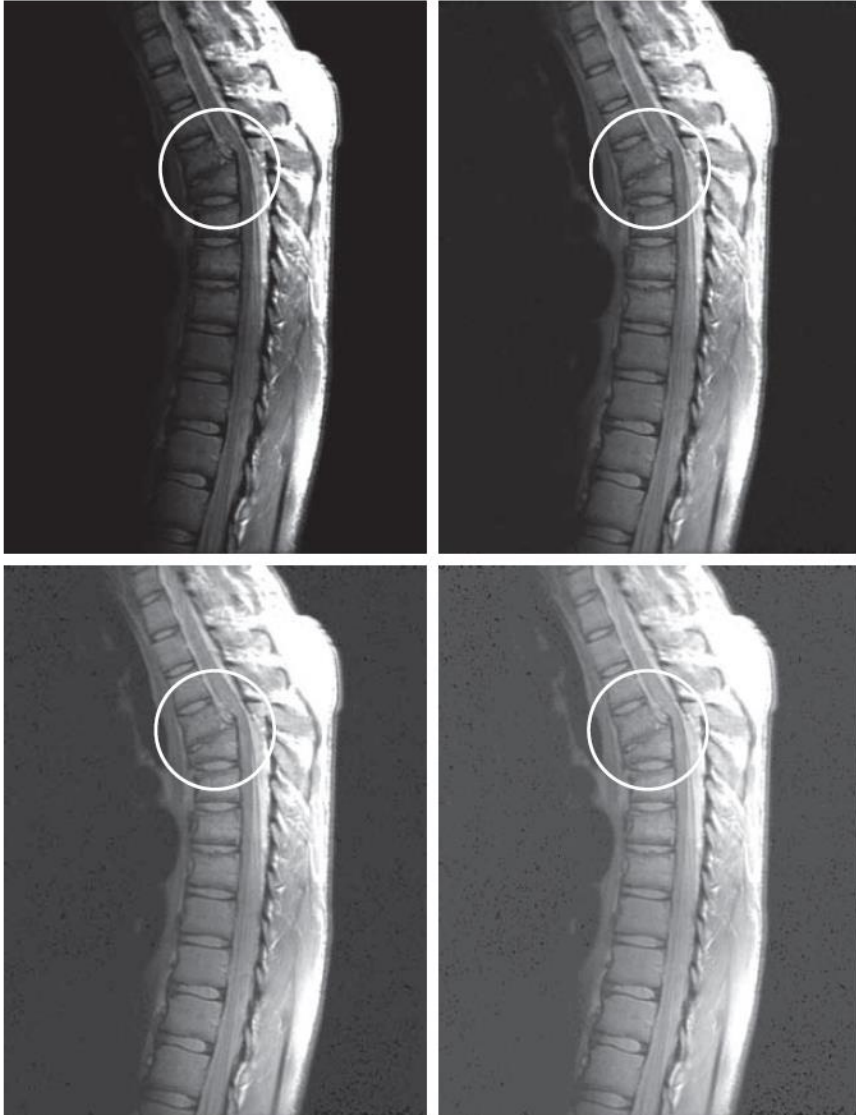
Power-law (Gamma) Transformations

a	b
c	d

FIGURE 3.8

(a) Magnetic resonance image (MRI) of a fractured human spine (the region of the fracture is enclosed by the circle).

(b)–(d) Results of applying the transformation in Eq. (3-5) with $c = 1$ and $\gamma = 0.6, 0.4,$ and 0.3 , respectively. (Original image courtesy of Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)



a	b
c	d

FIGURE 3.9

(a) Aerial image.

(b)–(d) Results of applying the transformation in Eq. (3-5) with $\gamma = 3.0, 4.0,$ and 5.0 , respectively. ($c = 1$ in all cases.) (Original image courtesy of NASA.)



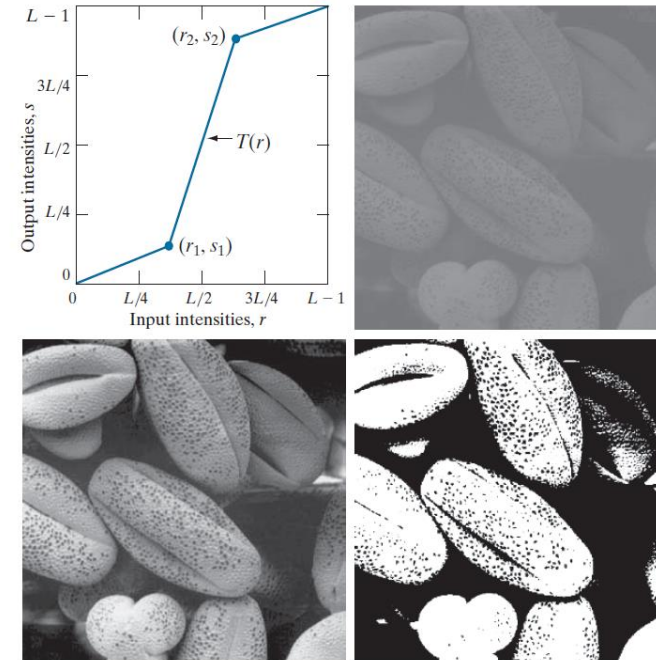
Some Basic Intensity Transformation Functions

- Contrast Stretching

a b
c d

FIGURE 3.10

Contrast stretching.
(a) Piecewise linear transformation function.
(b) A low-contrast electron microscope image of pollen, magnified 700 times.
(c) Result of contrast stretching.
(d) Result of thresholding.
(Original image courtesy of Dr. Roger Heady, Research School of Biological Sciences, Australian National University, Canberra, Australia.)

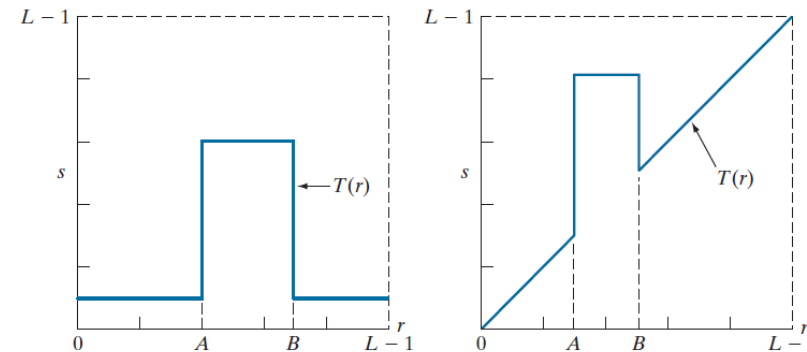


- Intensity-Level Slicing

a b

FIGURE 3.11

(a) This transformation function highlights range $[A, B]$ and reduces all other intensities to a lower level.
(b) This function highlights range $[A, B]$ and leaves other intensities unchanged.

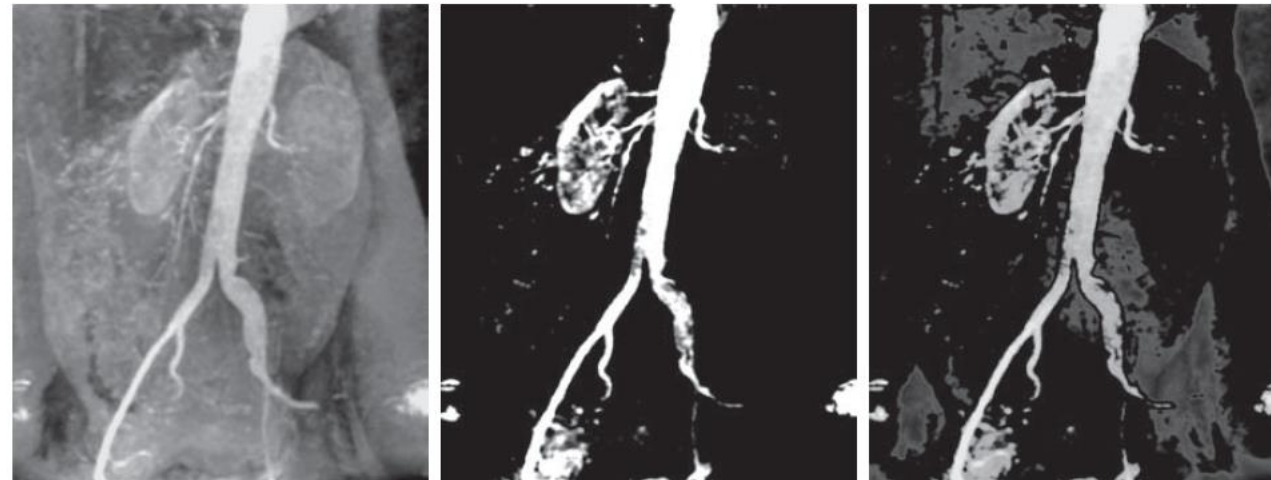
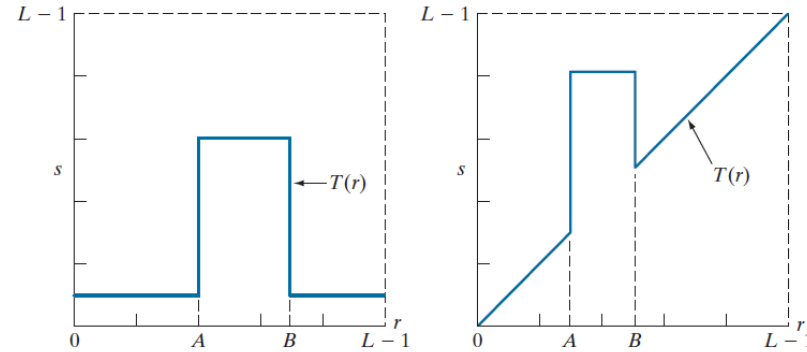


Intensity-Level Slicing

a b

FIGURE 3.11

(a) This transformation function highlights range $[A, B]$ and reduces all other intensities to a lower level.
(b) This function highlights range $[A, B]$ and leaves other intensities unchanged.



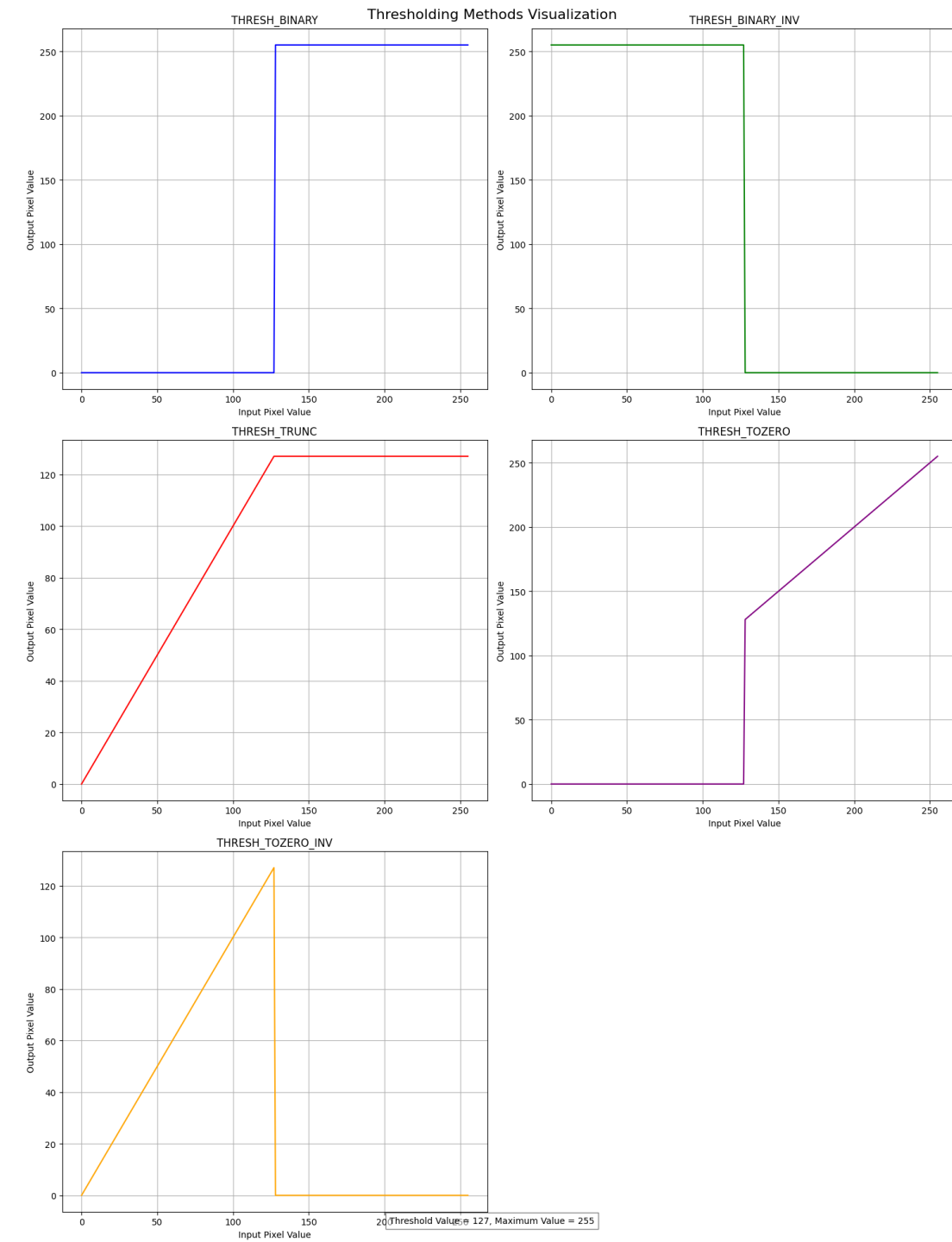
a b c

FIGURE 3.12 (a) Aortic angiogram. (b) Result of using a slicing transformation of the type illustrated in Fig. 3.11(a), with the range of intensities of interest selected in the upper end of the gray scale. (c) Result of using the transformation in Fig. 3.11(b), with the selected range set near black, so that the grays in the area of the blood vessels and kidneys were preserved. (Original image courtesy of Dr. Thomas R. Gest, University of Michigan Medical School.)

Intensity-Level Slicing

OpenCV's implementation:

- **cv2.THRESH_BINARY** Input Used for: Creating clear black and white images, separating objects from background
pixel > threshold $\rightarrow$ output = maxVal
Otherwise $\rightarrow$ output = 0
- **cv2.THRESH_BINARY_INV** Input Used for: Same as BINARY but inverted, useful when objects are darker than background
pixel > threshold $\rightarrow$ output = 0
Otherwise $\rightarrow$ output = maxVal
- **cv2.THRESH_TRUNC** Input Used for: Limiting pixel intensities to threshold value, preserving detail below threshold
pixel > threshold $\rightarrow$ output = threshold
Otherwise $\rightarrow$ output = original
- **cv2.THRESH_TOZERO** Input Used for: Keeping pixels above threshold unchanged, removing lower intensities
pixel > threshold $\rightarrow$ output = original
Otherwise $\rightarrow$ output = 0
- **cv2.THRESH_TOZERO_INV** Input Used for: Keeping pixels below threshold unchanged, removing higher intensities
pixel > threshold $\rightarrow$ output = 0
Otherwise $\rightarrow$ output = original



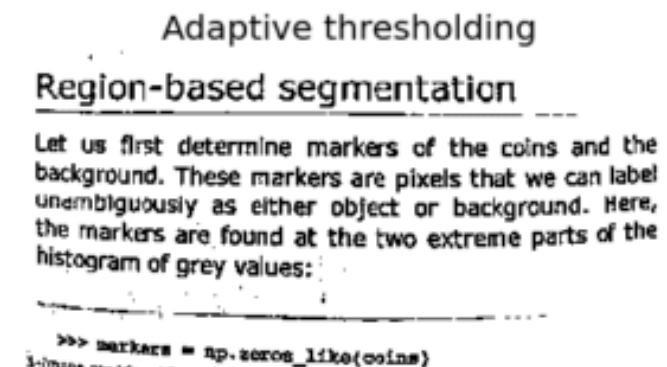
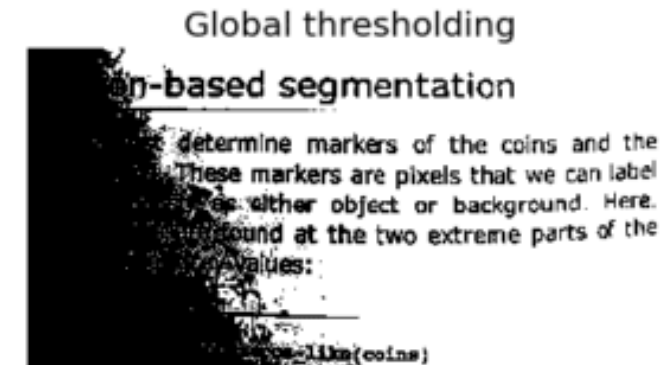
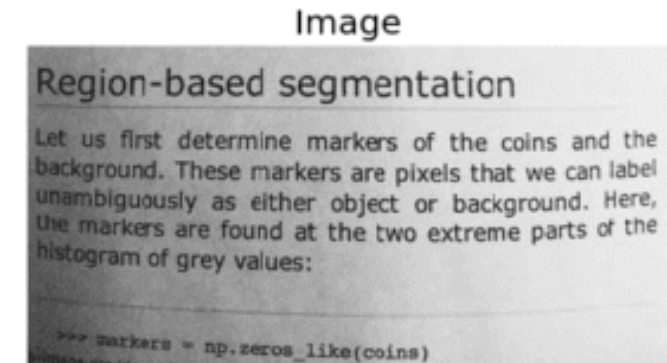
Adaptive Image Thresholding

Why Adaptive Thresholding

- Handles images with varying illumination
- Calculates threshold for each pixel based on surrounding region
- More robust than global thresholding
- Ideal for documents, natural scenes

Use Cases

- Document scanning
- OCR preprocessing
- License plate detection
- Barcode/QR code reading
- Medical image segmentation



Mean of neighborhood

```
thresh = cv.adaptiveThreshold(img, maxValue, cv.ADAPTIVE_THRESH_MEAN_C,  
                             cv.THRESH_BINARY, blockSize, C)
```

- Uses mean of neighborhood area
- blockSize: size of pixel neighborhood (odd number)
- C: constant subtracted from $\text{meanT}(x,y) = \text{mean}(\text{blockSize} \times \text{blockSize}) - C$

Weighted (Gaussian) Mean of neighborhood

```
thresh = cv.adaptiveThreshold(img, maxValue, cv.ADAPTIVE_THRESH_GAUSSIAN_C,  
                             cv.THRESH_BINARY, blockSize, C)
```

- Uses weighted sum (Gaussian window)
- Gives more weight to closer pixels
- $T(x,y) = \text{GaussianMean}(\text{blockSize} \times \text{blockSize}) - C$
- Better for gradual intensity changes

Parameter Selection

- blockSize: Usually 11, 15, or larger odd numbers
- C: Fine-tuning parameter (2-20 typical)
- maxValue: Usually 255 for 8-bit images



GEOMETRIC TRANSFORMATIONS

Geometric transformations

1. Spatial transformation of coordinates.

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \mathbf{T} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

transformation of coordinates

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \mathbf{A} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Affine transformations

2. [Intensity interpolation](#) that assigns intensity values to the spatially transformed pixels.

TABLE 2.3
Affine
transformations
based on
Eq. (2-45).

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \mathbf{A} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Affine transformations

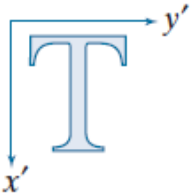
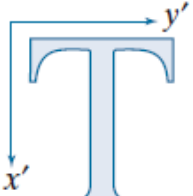
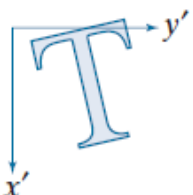
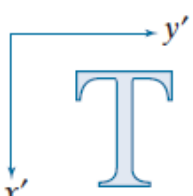
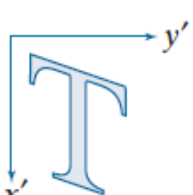
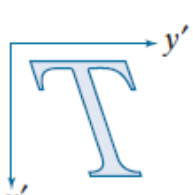
Transformation Name	Affine Matrix, A	Coordinate Equations	Example
Identity	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x$ $y' = y$	
Scaling/Reflection (For reflection, set one scaling factor to -1 and the other to 0)	$\begin{bmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = c_x x$ $y' = c_y y$	
Rotation (about the origin)	$\begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x \cos \theta - y \sin \theta$ $y' = x \sin \theta + y \cos \theta$	
Translation	$\begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x + t_x$ $y' = y + t_y$	
Shear (vertical)	$\begin{bmatrix} 1 & s_v & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x + s_v y$ $y' = y$	
Shear (horizontal)	$\begin{bmatrix} 1 & 0 & 0 \\ s_h & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x$ $y' = s_h x + y$	

Image Histogram

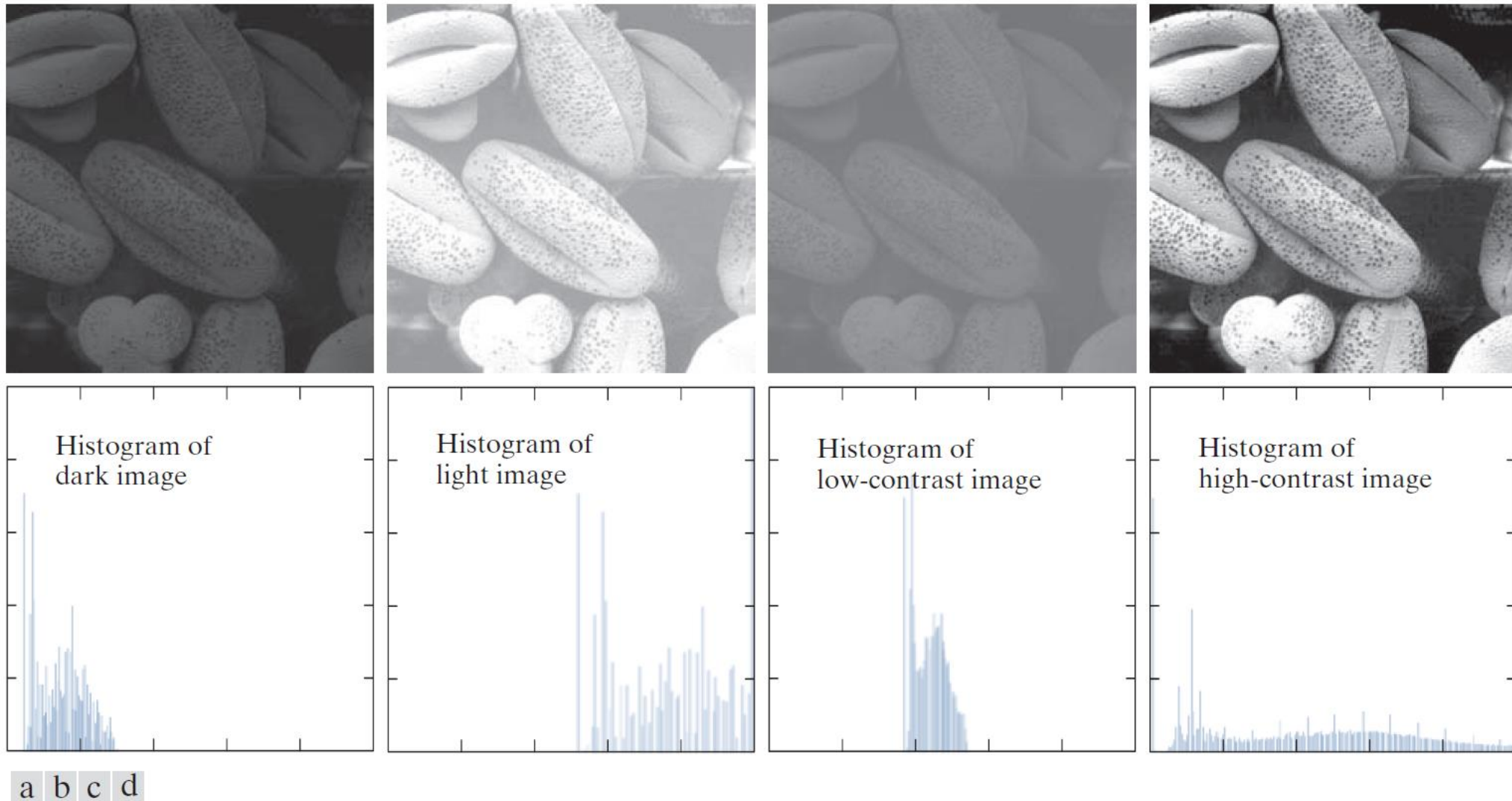


FIGURE 3.16 Four image types and their corresponding histograms. (a) dark; (b) light; (c) low contrast; (d) high contrast. The horizontal axis of the histograms are values of r_k and the vertical axis are values of $p(r_k)$.

Histogram equalization (optional)

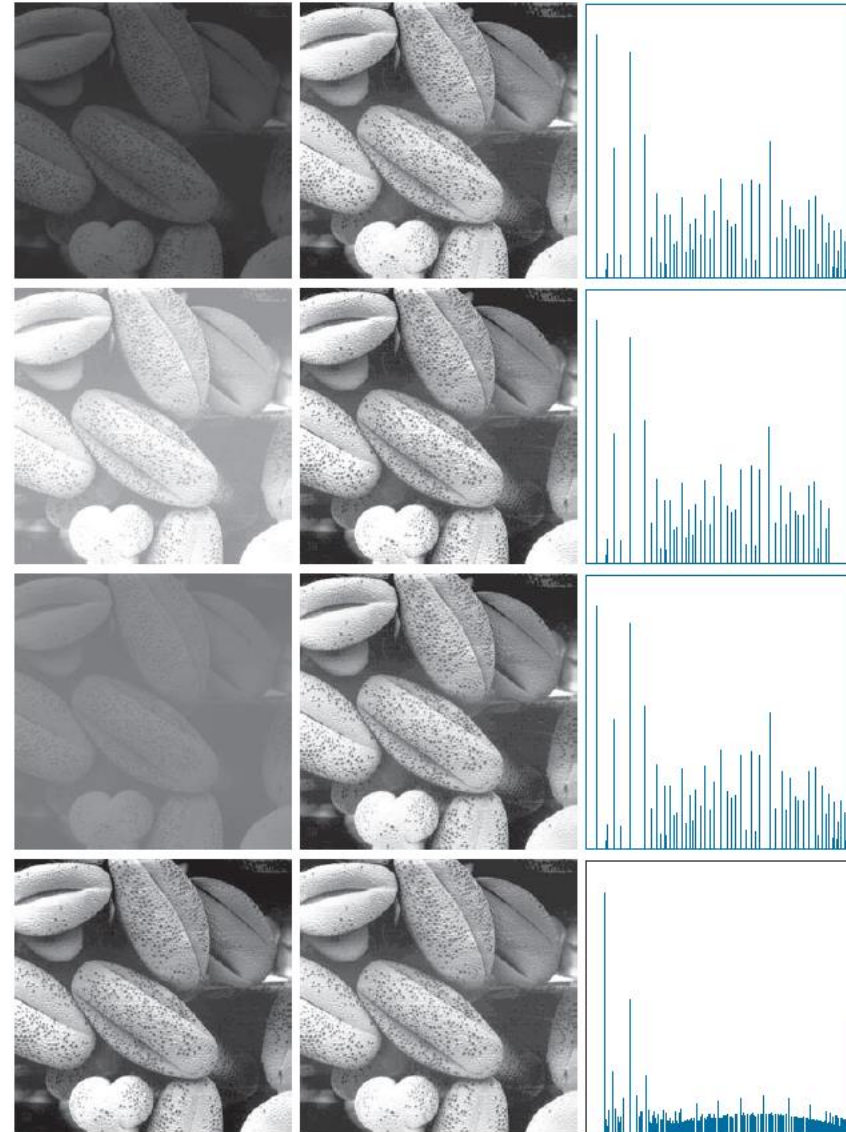


FIGURE 3.20 Left column: Images from Fig. 3.16. Center column: Corresponding histogram-equalized images. Right column: histograms of the images in the center column (compare with the histograms in Fig. 3.16).